DATAWAREHOUSE

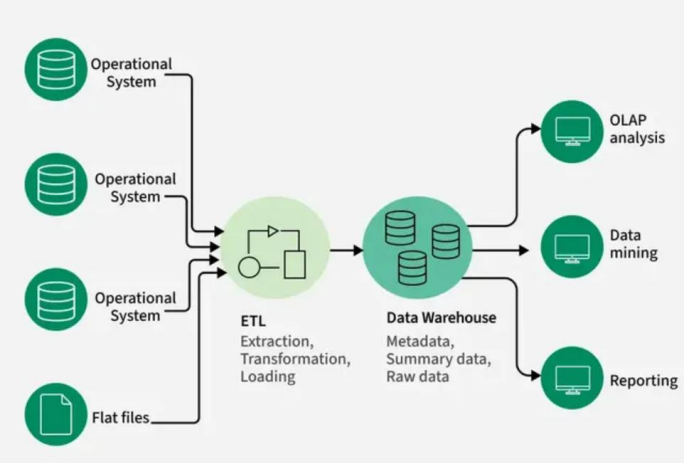
Data and Information?

- Data is an individual unit that contains raw materials which do not carry any specific meaning.
- Information is a group of data that collectively carries a logical meaning.
- Data doesn't depend on information. Information depends on data.
- Data is measured in bits and bytes. Information is measured in meaningful units like time, quantity, etc.

Data Warehouse:

A data warehouse is a centralized system used for storing and managing large volumes of data from various sources. It is designed to help businesses analyze historical data and make informed decisions. Data from various systems is gathered, cleaned, and arranged properly so it can be easily searched and used for reports.

- Goal is to produce statistical results that may help in decision-making.
- Ensures fast data retrieval even with the vast datasets.



Need for Data Warehousing

1. **Managing Huge Data Easily**
   Normal databases can only hold a limited amount of data, but data warehouses are made to store very large amounts (in terabytes). This helps companies keep years of data safely.
2. **Better Data Analysis**
   Regular databases are meant for daily tasks, not for deep analysis. Data warehouses are built for analysis, making it easier to run complex reports and get useful insights.
3. **One Place for All Data**
   A data warehouse collects data from many sources and stores it in one place. This gives businesses a complete and clear view of their operations.
4. **Finding Patterns and Trends**
   By storing past data, companies can study changes over time. This helps them make smarter decisions and plan for the future.
5. **Helps in Smarter Business Decisions**
   Data warehouses work with tools that help in reporting and analysis. This gives managers quick access to important information, improving decision-making and business performance.

Components of Data Warehouse

The main components of a data warehouse include:

- **Data Sources**: These are the original places where data comes from, like company databases or outside data providers.
- **ETL (Extract, Transform, Load)**: This is the process of taking data from different sources, cleaning and converting it, and then putting it into the data warehouse.
- **Data Warehouse Database**: This is the main storage area where all cleaned data is kept. It is arranged in a way that makes it easy to search and analyze.
- **Metadata**: This means "data about data." It explains where the data came from, what it means, and how to use it.
- **Data Marts**: These are smaller parts of the warehouse, focused on a specific area like sales or marketing, and used by certain departments.
- **OLAP Tools**: These tools help users explore data from different angles to find trends or patterns.
- **End-User Access Tools**: These include dashboards and Business Intelligence (BI) tools that let users create reports and view data easily.

Characteristics of Data Warehousing

Data warehousing is very useful for handling and understanding large amounts of data. It helps organizations gather all their data in one place so they can study it and make better decisions.

1. Central Storage of Data:

A data warehouse keeps all the data from different places (like business systems, databases, or external files) in one central location. This helps businesses see all their data clearly and make smart choices.

2. Combining Data from Different Sources:

It brings together data from many different systems into a single, organized format. This avoids confusion caused by scattered or mismatched data.

3. Storing Old (Historical) Data:

A data warehouse saves past data too. This helps in checking how things have changed over time and in spotting trends or problems.

4. Easy Data Search and Analysis:

It allows users to quickly search and analyze the data. This helps them find useful patterns or trends, leading to better decisions.

5. Data Cleaning and Formatting:

Before storing, the data is cleaned and formatted. This process removes errors and makes sure all the data follows the same structure.

6. Discovering Patterns (Data Mining):

With data mining tools, businesses can find hidden patterns or links in the data. This can help them plan for the future or find new business ideas.

7. Keeping Data Safe:

A data warehouse also keeps data safe using methods like:

- Access control (who can see the data),
- Encryption (protecting data from hackers),
- Backups (copies of data in case of loss).

Types of Data Warehouses

The different types of Data Warehouses are:

- **Enterprise Data Warehouse (EDW):** A centralized warehouse that stores data from across the organization for analysis and reporting.
- **Operational Data Store (ODS):** Stores real-time operational data used for day-to-day operations, not for deep analytics.
- **Data Mart:** A subset of a data warehouse, focusing on a specific business area or department.
- **Cloud Data Warehouse:** A data warehouse hosted in the cloud, offering scalability and flexibility.
- **Big Data Warehouse:** Designed to store vast amounts of unstructured and structured data for big data analysis.
- **Virtual Data Warehouse:** Provides access to data from multiple sources without physically storing it.
- **Hybrid Data Warehouse:** Combines on-premises and cloud-based storage to offer flexibility.
- **Real-time Data Warehouse:** Designed to handle real-time data streaming and analysis for immediate insights.

Architecture of Data Warehousing:

Tier-1:

The bottom tier is a warehouse database server that is almost always a relational database system. Back-end tools and utilities are used to feed data into the bottom tier from operational databases or other external sources (such as customer profile information provided by external consultants). These tools and utilities perform data extraction, cleaning, and transformation (e.g., to merge similar data from different sources into a unified format), as well as load and refresh functions to update the data warehouse. The data are extracted using application program interfaces known as gateways. A gateway is supported by the underlying DBMS and allows client programs to generate SQL code to be executed at a server.

Examples of gateways include ODBC (Open Database Connection) and OLEDB (Open Linking and Embedding for Databases) by Microsoft and JDBC (Java Database Connection). This tier also contains a metadata repository, which stores information about the data warehouse and its contents.

Tier-2:

The middle tier is an OLAP server that is typically implemented using either a relational OLAP (ROLAP) model or a multidimensional OLAP. OLAP model is an extended relational DBMS that maps operations on multidimensional data to standard relational operations. A multidimensional OLAP (MOLAP) model, that is, a special-purpose server that directly implements multidimensional data and operations.

Tier-3:

The top tier is a front-end client layer, which contains query and reporting tools, analysis tools, and/or data mining tools (e.g., trend analysis, prediction, and so on).

OR

A Data Warehouse has a 3-layer architecture −

Data Source Layer:

It defines how the data comes to a Data Warehouse. It involves various data sources and operational transaction systems, flat files, applications, etc.

Integration Layer:

It consists of Operational Data Store and Staging area. Staging area is used to perform data cleansing, data transformation and loading data from different sources to a data warehouse. As multiple data sources are available for extraction at different time zones, staging area is used to store the data and later to apply transformations on data.

Presentation Layer:

This is used to perform BI reporting by end users. The data in a DW system is accessed by BI users and used for reporting and analysis.

The following illustration shows the common architecture of a Data Warehouse System.

OLAP (Online Analytical Processing):

OLAP is a technology used to quickly answer **multi-dimensional analytical queries**. It helps users **analyse large volumes of data** in different ways to support **business decision-making**.

In simple words:
OLAP lets users look at data from different angles (dimensions) – like time, location, product – and find useful patterns or summaries.

Characteristics of OLAP

1. **Multidimensional View of Data**
   - Data is viewed in the form of a **cube** with multiple **dimensions**.
   - For example, sales data can be viewed by **product**, **region**, and **time**.
2. **Fast Query Performance**
   - Designed for **quick analysis** even when data is huge.
3. **Supports Complex Calculations**
   - Allows mathematical functions like **totals, averages, trends**.
4. **Interactive Analysis**
   - Users can perform operations like **drill-down**, **roll-up**, **slice**, and **dice** (explained below).

Operations Of OLAP:

Operation	Description	Example
Roll-up	Summarizes data to a higher level	Monthly sales → Yearly sales
Drill-down	Goes deeper into more detailed data	Yearly sales → Monthly sales
Slice	Fixes one dimension and views the rest	View sales only for 2024
Dice	Views data for selected values from multiple dimensions	Sales for 2024, in Delhi, for Product A
Pivot (Rotate)	Rotates data to see different perspectives	Swap rows and columns in a report

Examples of Operation

1. Sales data is recorded daily. Rolling up could mean seeing it monthly or yearly instead of daily.
2. If you have **monthly sales**, you can drill down to see **daily sales** for January.
3. From a cube with **Product, Region, Time**, you slice by **Time = 2024**, and view only 2024 data for all products and regions.
4. Show sales for **Products A and B**, in **Regions East and North**, during **Q1 and Q2**.
5. Swap between viewing:

   Products in rows and Regions in columns
   vs.
   Regions in rows and Products in columns

Needs of OLAP:

- Helps in **decision making**.
- Identifies **trends, patterns, and outliers**.
- Supports **business intelligence** and **reporting**.
- Saves time with fast, interactive queries.
- Allows **what-if analysis** for future planning.

Type of OLAP servers:

The three major types of OLAP servers are as follows:

- **ROLAP**
- **MOLAP**
- **HOLAP**

1. Relational OLAP (ROLAP):

ROLAP stands for **Relational Online Analytical Processing**.
It works with data stored in **relational databases** (like MySQL, Oracle, etc.). Both the main data and supporting tables (called dimension tables) are stored in a **table format**.

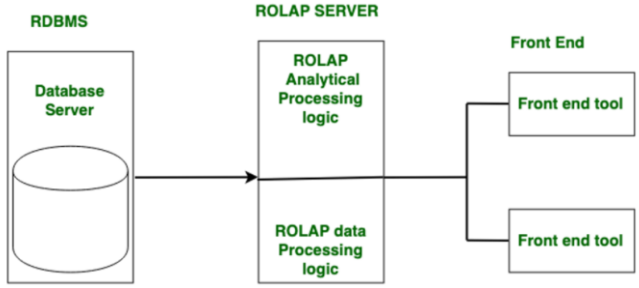
A **ROLAP server** connects the database (back-end) to the user tools (front-end), helping users run analytical queries. It uses **RDBMS (Relational Database Management System)** to store and manage data, while special OLAP tools handle analysis.

Advantages of ROLAP:

- Works well with existing **data warehouses** and **transaction systems (OLTP)**.
- Can handle **very large amounts of data**, depending on the capacity of the database system used. There's **no fixed size limit** from ROLAP itself.

Disadvantages of ROLAP:

- It relies on **SQL**, so if SQL is limited, analysis options are also limited.
- It's **hard to keep summary tables updated**, especially when the data changes frequently.



2. **Multidimensional OLAP (MOLAP):**

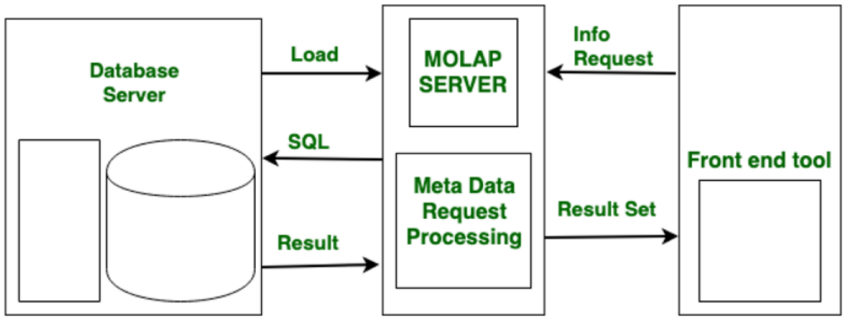
   MOLAP stands for **Multidimensional Online Analytical Processing**.
   It stores **data in the form of cubes**, not in tables like ROLAP. These cubes are built before the user runs queries, so results come **very fast**.
   The data is stored in a **multidimensional format**, which makes it easy to **perform complex calculations and quick analysis**.

Advantages of MOLAP:

- **Very fast** for reading data because everything is pre-calculated.
- **Better performance** for complex queries.
- **Uses less memory** by compressing data.

Disadvantages of MOLAP:

- **Not suitable for very large data sets** — building cubes takes time and storage.
- Limited to the amount of data the cube can hold.
- **Pre-processing required**, so it takes time to update new data.



3. **Hybrid OLAP (HOLAP):**

HOLAP stands for **Hybrid Online Analytical Processing**.
It is a **combination of MOLAP and ROLAP**, using the strengths of both.

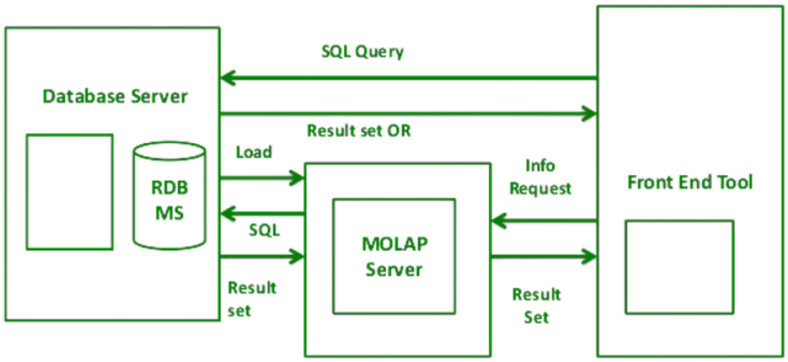
- It stores **detailed data in relational databases (like ROLAP)**
- It stores **summary data in cubes (like MOLAP)**
  This makes it **faster than ROLAP** and **more scalable than MOLAP**.

Advantages of HOLAP:

- **Balances performance and storage**.
- Can handle **large amounts of data**.
- Uses **cubes for quick summary results**, and **relational tables for detailed data**.

Disadvantages of HOLAP:

- **More complex system** to manage.
- May require **advanced configuration and tools**.



Online Transaction Processing (OLTP):

OLTP stands for **Online Transaction Processing**.
It is used to **manage day-to-day operations** of businesses like **banking, retail, airline booking**, etc.
It handles **frequent, small transactions** such as insert, update, and delete.
OLTP systems are **designed for speed and accuracy** in recording real-time transactions.

Characteristics of OLTP:

- Manages **real-time data** (current/up-to-date information).
- Supports **short and simple queries**.
- Allows **fast inserts, updates, and deletions**.
- Focuses on **data integrity and accuracy**.
- Supports **many users at the same time** (concurrent access).

Examples of OLTP Systems:

- **ATM transactions**
- **Online shopping checkout**
- **Railway or flight booking system**
- **Banking systems**

Advantages of OLTP:

- **High speed** for processing daily operations.
- **Real-time data access** and updates.
- Supports **large numbers of users** simultaneously.
- Ensures **data accuracy and consistency** using transactions.

Disadvantages of OLTP:

- **Not suitable for complex analysis or reporting**.
- Can slow down if **too many complex queries** are run.
- Focuses only on **current data**, not historical trends.

Data Warehouse Models

A **Data Warehouse Model** describes how data is organized, stored, and managed inside the data warehouse.

It helps in understanding **how data flows**, how it is structured, and how users interact with it.

There are **three main types of data warehouse models**:

1. Enterprise Data Warehouse (EDW)

- A **centralized warehouse** that stores data for the **entire organization**.
- It is a **single, consistent data source** for decision-making across departments (HR, Finance, Sales, etc.).

Features:

- Stores **historical and current data**.
- Supports **complex queries and reporting**.
- **Well-structured and secure**.

Example:

A company with multiple branches uses one enterprise data warehouse to store and analyze sales, employee, and customer data from all branches.

2. Data Mart

- A **smaller version** of a data warehouse.
- Focused on a **specific department or function** (like Sales, HR, Finance).

Features:

- Easier and faster to implement.
- Stores **data relevant to one subject** or business unit.
- Can be **independent or linked** to the main warehouse.

Types:

- **Dependent Data Mart**: Gets data from EDW.
- **Independent Data Mart**: Has its own source systems.

Example:

A sales data mart stores only sales data for use by the sales department.

3. Operational Data Store (ODS)

- A **temporary storage area** used to collect and clean data **before loading** it into the data warehouse.
- Stores **real-time or near-real-time data**.

Features:

- Used for **daily operations** and **short-term analysis**.
- Updated **frequently**.
- Acts as a **bridge** between operational systems and the data warehouse.

Example:

Banking systems use ODS to store live transaction data before updating the main warehouse.

Metadata Repository:

A Metadata Repository is a central place where all metadata is stored and managed. It acts like a dictionary or guidebook that explains everything about the data in the warehouse. It helps users, developers, and tools understand:

- Where the data came from
- How the data is organized
- What the data means
- How to access and use the data

Components of a Metadata Repository

1. Storing Metadata:
These **repositories** hold all the info about our **datasets**. This includes where the data comes from, how it's set up, and any changes it's been through.

2. Guiding Users:
When someone needs specific data, the **repository** guides them to the **right** files. Users can see which files are there, when they were added, and who put them there.

3. Tracking Data Changes:
These repositories also keep **track** of any changes to the data. So if a dataset gets **updated** or changed in any way, the repository shows this, making sure everyone always has the most **accurate** data.

Benefits of a Metadata Repository

1. Organization:
A **metadata** repository organizes datasets in one place, making it easier to search and find what you need. Instead of sifting through vast amounts of files, users can quickly locate data based on its metadata, improving **workflow** and **efficiency**.

2. Governance:
They help **organizations** keep an eye on how data is being used. You can see who has been using or changing the data and making sure everything is done right.

3. Efficiency:
By **storing** detailed information about our data, these **repositories** stop us from doing the same work over and over again. We can easily check if the data we need is already there saving time and space by not adding or **downloading** the same stuff again and again.

Uses of a Metadata Repository

- **Centralized Data Management:** Consolidating all dataset information into one system streamlines the management and retrieval of data.
- **Tracking Data History:** Metadata repositories track when and how datasets are **modified**, providing a **historical** record of changes.
- **Facilitating Compliance and Governance:** They help organizations maintain control over data usage, ensuring **compliance** with regulations and governance policies.
- **Enhancing User Accessibility to Datasets:** Metadata repositories improve data **accessibility**, allowing users to quickly find and access the data they need.

Features of a Metadata Repository

- **Data Storage and Organization:** A well-structured repository makes it easy to **categorize** and manage datasets.
- **Tracking of Data Origin and Modifications:** Each dataset's journey is **logged**, helping users trace its origin and any modifications made.
- **User Access and Permission Management:** Repositories **control** who can access or edit data, ensuring **security** and proper data handling.
- **Historical Data Tracking:** Detailed records of dataset **updates** and **changes** allow users to go back and understand how data has evolved.

Schema in Data Warehouse

A **schema** is a **blueprint** that defines how data is organized in a database. In data warehousing, it shows how **fact tables** and **dimension tables** are linked.

Common Schemas in Multidimensional Models

There are **three main schemas** used to represent data in multidimensional format:

1. Star Schema

- The most **simple and popular** schema.
- Has **one central fact table** surrounded by multiple **dimension tables** — looks like a star.

Features:

- **Fact table**: Contains numeric measures (e.g., Sales, Profit).
- **Dimension tables**: Contain descriptive data (e.g., Product, Customer, Time, Location).
- Easy to understand and use in queries.

Pros:

- Fast and easy for query and reporting.
- Easy to design.

Cons:

- Can have **redundant data** in dimension tables.

2. Snowflake Schema

- An **extension of star schema** where dimension tables are **normalized** (split into sub-tables).
- Looks like a **snowflake shape**.

Features:

- Breaks down large dimension tables into smaller ones.
- Reduces data redundancy.

Pros:

- **Less redundancy**, saves storage.
- Good for handling **complex dimensions**.

Cons:

- Slower query performance due to more joins.
- Harder to understand.

3. Fact Constellation Schema (Galaxy Schema)

- Has **multiple fact tables** that share **common dimension tables**.
- Also known as a **Galaxy Schema**.

Features:

- Suitable for **complex applications** (e.g., e-commerce, banking).
- Reuses dimension tables for multiple business areas.

Pros:

- Supports **multiple subject areas** (sales, purchase, etc.)
- **Flexible and reusable** design.

Cons:

- Complex structure.
- Hard to design and maintain.